

Math 550, S21
Exam 2

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	20	
2	22	
3	22	
4	22	
5	20	
Total	106	

Good Luck !

Problem 1. (20 points)

(a) Evaluate $\int_0^1 \int_0^{x^2} \frac{y}{x^5 + 1} dy dx$

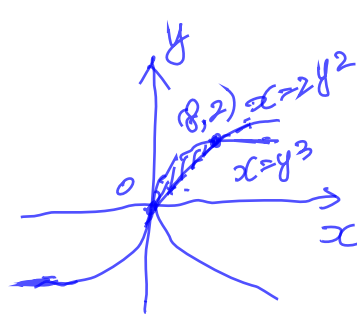
$$\begin{aligned} &= \int_0^1 \frac{1}{x^5 + 1} \left[\frac{y^2}{2} \right]_{y=0}^{y=x^2} dx \\ &= \frac{1}{2} \int_0^1 \frac{x^4}{x^5 + 1} dx \\ &= \left[\frac{1}{10} \ln |x^5 + 1| \right]_{x=0}^{x=1} \\ &= \frac{1}{10} [\ln 2 - \ln 1] = \frac{1}{10} \ln 2 \end{aligned}$$

(b) Evaluate $\int_0^1 \int_0^z \int_0^y z e^{y^2} dx dy dz$

$$\begin{aligned} &= \int_0^1 \int_0^z z e^{y^2} [x]_{x=0}^{x=y} dy dz \\ &= \int_0^1 \int_0^z z y e^{y^2} dy dz \\ &= \int_0^1 \frac{1}{2} z [e^{y^2}]_{y=0}^{y=z} dz \\ &= \int_0^1 \frac{z}{2} e^{z^2} - \frac{z}{2} dz \\ &= \left[\frac{1}{4} e^{z^2} - \frac{z^2}{4} \right]_{z=0}^{z=1} \\ &= \frac{1}{4} [e - 1] - \frac{1}{4} [1] \\ &= \frac{1}{4} (e - 2) \end{aligned}$$

Problem 2. (22 points)

(a) Let D be the region in $\mathbb{R}^2$ bounded by $x = 2y^2$ and $x = y^3$. Evaluate $\iint_D 2x + y \, dx \, dy$.



$$= y^2 \begin{cases} x = 2y^2 \\ x = y^3 \end{cases} \Rightarrow y^3 - 2y^2 = 0, y^2(y-2) = 0$$

$$\Rightarrow y = 0, y = 2$$

$$(x=0) \quad (x=8)$$

$$D = \{(x, y) \mid 0 \leq y \leq 2, y^3 \leq x \leq 2y^2\}$$

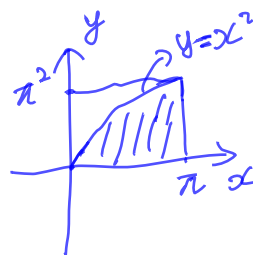
$$\iint_D 2x + y \, dx \, dy = \int_0^2 \int_{y^3}^{2y^2} 2x + y \, dx \, dy$$

$$= \int_0^2 [x^2 + xy]_{x=y^3}^{x=2y^2} dy = \int_0^2 (-y^6 + 3y^4 + 2y^3) dy$$

$$= \left[-\frac{y^7}{7} + \frac{3}{5}y^5 + \frac{1}{2}y^4 \right]_{y=0}^{y=2} = -\frac{128}{7} + \frac{96}{5} + 8$$

$$= \frac{912}{35}$$

(b) Evaluate $\int_0^{\pi^2} \int_{\sqrt{y}}^{\pi} \frac{\sin x}{x} \, dx \, dy$ by reversing the order of the integration.



$$D = \{(x, y) \mid 0 \leq y \leq \pi^2, \sqrt{y} \leq x \leq \pi\}$$

$$= \{(x, y) \mid 0 \leq x \leq \pi, 0 \leq y \leq x^2\}$$

Thus

$$\int_0^{\pi^2} \int_{\sqrt{y}}^{\pi} \frac{\sin x}{x} \, dx \, dy = \int_0^{\pi} \int_0^{x^2} \frac{\sin x}{x} \, dy \, dx$$

$$= \int_0^{\pi} \left(\frac{\sin x}{x} \right) (x^2 - 0) \, dx$$

$$= \int_0^{\pi} x \sin x \, dx$$

$$= [-x \cos x]_{x=0}^{x=\pi} - \int_0^{\pi} (-\cos x) \, dx$$

$$= \pi + [\sin x]_{x=0}^{x=\pi} = \pi$$

Problem 3. (22 points)

(a) Let W be the hemisphere defined by $x^2 + y^2 + z^2 \leq 9$, $z \geq 0$ with a density function $f(x, y, z) = z$. Find the mass of the hemisphere H by evaluating $\iiint_W z \, dx \, dy \, dz$ using the triple integral in **spherical coordinates**.

$$\begin{aligned}
 W &= \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 9, z \geq 0\} \\
 \Rightarrow W^* &= \{(\rho, \phi, \theta) \mid 0 \leq \rho \leq 3, 0 \leq \phi \leq \frac{\pi}{2}, 0 \leq \theta \leq 2\pi\} \\
 \iiint_W z \, dx \, dy \, dz &= \iiint_{W^*} \rho \sin \phi \, d\rho \, d\phi \, d\theta \\
 &= \int_0^3 \int_0^{\pi/2} \int_0^{2\pi} (\rho \sin \phi) \rho^2 \sin \phi \, d\theta \, d\phi \, d\rho \\
 &= 2\pi \int_0^3 \int_0^{\pi/2} \rho^3 \sin^2 \phi \, d\phi \, d\rho \\
 &= 2\pi \int_0^3 \rho^3 \left[\frac{\sin^2 \phi}{2} \right]_{\phi=0}^{\phi=\pi/2} d\rho \\
 &= \pi \int_0^3 \rho^3 d\rho = \pi \cdot \left[\frac{\rho^4}{4} \right]_{\rho=0}^{\rho=3} \\
 &= \frac{81}{4} \pi
 \end{aligned}$$

(b) Evaluate the iterated triple integral $\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{\sqrt{x^2+y^2}} e^z \, dz \, dy \, dx$ by using **cylindrical coordinates**.

$$\begin{aligned}
 W &= \{(x, y, z) \mid -2 \leq x \leq 2, -\sqrt{4-x^2} \leq y \leq \sqrt{4-x^2}, 0 \leq z \leq \sqrt{x^2+y^2}\} \\
 (r, \theta, z) &\Downarrow x = r \cos \theta, y = r \sin \theta, z = z \\
 W^* &= \{(r, \theta, z) \mid 0 \leq r \leq 2, 0 \leq \theta \leq 2\pi, 0 \leq z \leq r\} \\
 \iiint_W e^z \, dz \, dy \, dx &= \iiint_{W^*} e^z r \, dr \, d\theta \, dz \\
 &= \int_0^2 \int_0^{2\pi} \int_0^r e^z r \, dz \, d\theta \, dr \\
 &= \int_0^2 \int_0^{2\pi} r [e^z]_{z=0}^{z=r} d\theta \, dr = \int_0^2 \int_0^{2\pi} r(e^r - 1) d\theta \, dr \\
 &= 2\pi \int_0^2 r(e^r - 1) dr = 2\pi \left[r(e^r - 1) \Big|_{r=0}^{r=2} - \int_0^2 e^r r \, dr \right] \\
 &= 2\pi \left[2(e^2 - 1) - (e^r - \frac{r^2}{2}) \Big|_{r=0}^{r=2} \right] \\
 &= 2\pi(e^2 - 1)
 \end{aligned}$$

Problem 4. (22 points)

(a) Evaluate the scalar line integral $\int_{\mathbf{x}} yz \cos x \, ds$ where the path $\mathbf{x}(t) = (t, 3 \cos t, 3 \sin t)$ with $0 \leq t \leq \pi$.

$$\begin{aligned} \mathbf{x}'(t) &= (1, -3\sin t, 3\cos t) \\ \Rightarrow \|\mathbf{x}'(t)\| &= \sqrt{1^2 + (-3\sin t)^2 + (3\cos t)^2} = \sqrt{10} \\ \int_{\mathbf{x}} yz \cos x \, ds &= \int_0^\pi (3\cos t)(3\sin t)(\cos t) \cdot \|\mathbf{x}'(t)\| \, dt \\ &= \sqrt{10} \int_0^\pi 9\sin t \cos^2 t \, dt \\ &= 9\sqrt{10} \left[-\frac{\cos^3 t}{3} \right]_{t=0}^{t=\pi} \\ &= 9\sqrt{10} \left(\frac{1}{3} + \frac{1}{3} \right) = 6\sqrt{10} \end{aligned}$$

(b) Evaluate the vector line integral $\int_C xy \, dx + (xy^3 - x) \, dy$ where C is the oriented curve with equation $x = y^2$, from $(1, 1)$ to $(4, -2)$.

$$\begin{aligned} \mathbf{x}(t): x &= t^2, y = -t, -1 \leq t \leq 2. \\ \mathbf{x}'(t) &= (2t, -1) \\ \int_C xy \, dx + (xy^3 - x) \, dy &= \int_{-1}^2 t^2(-t)(2t \, dt) + [t^2(-t)^3 - t^2](-dt) \\ &= \int_{-1}^2 -2t^4 + t^5 + t^2 \, dt \\ &= \left[-\frac{2}{5}t^5 + \frac{1}{6}t^6 + \frac{1}{3}t^3 \right]_{t=-1}^{t=2} \\ &= \left(-\frac{64}{5} + \frac{32}{3} + \frac{8}{3} \right) - \left(\frac{2}{5} + \frac{1}{6} - \frac{1}{3} \right) \\ &= \frac{3}{10} \end{aligned}$$

Problem 5. (20 points)

(a) Use **Green's theorem** to compute the line integral $\oint_C (y^2 - 2xy) dx + (x^2 + 2xy) dy$ (i.e. by converting it to a double integral), where C is the boundary of the square with vertices $(0, 0)$, $(2, 0)$, $(2, 2)$, and $(0, 2)$, oriented counterclockwise.

$$\begin{aligned} D &= \{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq 2\} \\ C &= \partial D. \\ \oint_C (y^2 - 2xy) dx + (x^2 + 2xy) dy &= \iint_D \left[\frac{\partial}{\partial x} (x^2 + 2xy) - \frac{\partial}{\partial y} (y^2 - 2xy) \right] dx dy \\ &= \int_0^2 \int_0^2 (2x + 2y - 2y + 2x) dx dy \\ &= \int_0^2 \int_0^2 4x dx dy = \int_0^2 [2x^2]_{x=0}^{x=2} dy \\ &= \int_0^2 8 dy = 8 \cdot 2 = 16 \end{aligned}$$

(b) Given a two-dimensional vector field $\mathbf{F}(x, y) = (xy^2)\mathbf{i} + (x^2y + y)\mathbf{j}$. Show that $\mathbf{F}$ is a conservative vector field and then find a potential function $f(x, y)$ such that $\mathbf{F} = \nabla f$.

$$\begin{aligned} M(x, y) &= xy^2, \quad N(x, y) = x^2y + y \\ \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} &= \frac{\partial}{\partial x} [x^2y + y] - \frac{\partial}{\partial y} [xy^2] = 2xy - 2xy = 0 \\ &\Rightarrow \mathbf{F} \text{ is conservative.} \\ \nabla f = \mathbf{F} &\Rightarrow \begin{cases} \frac{\partial f}{\partial x} = xy^2 & \text{--- ①} \\ \frac{\partial f}{\partial y} = x^2y + y & \text{--- ②} \end{cases} \\ \text{From ①, we get } f(x, y) &= \int xy^2 dx + g(y) \\ &= \frac{1}{2}xy^2 + g(y) & \text{--- ③} \\ \text{put ③ into ②, we get } \frac{\partial}{\partial y} \left[\frac{1}{2}xy^2 + g(y) \right] &= x^2y + y \\ \Rightarrow g'(y) = y &\Rightarrow g(y) = \int y dy = \frac{y^2}{2} + C \\ \text{Thus } f(x, y) &= \frac{1}{2}xy^2 + \frac{1}{2}y^2 + C \end{aligned}$$